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Executive Summary

Optical fingerprint modules for Arduino ("four-wire" units) include common models like Zhian's ZFM-20 series and AS608-based sensors (often sold by Adafruit and others) [1] [2]. These modules use TTL serial (UART) at 3.3–5 V and typically have wires colored Red (VCC), Black (GND), White and Green (RX/TX) [3] [1]. We tabulate wiring conventions (Red=VCC, Black=GND, White=RX, Green=TX) across popular modules. Optical sensors image a finger under illumination; onboard DSP hardware processes the ridge/valley pattern into a template and matches it against stored templates [4] [5]. A flowchart details the data flow from finger placement to match decision. Limitations include sensitivity to dirt, lighting and finger condition, spoofing (presentation attacks), false accept/reject errors, limited template memory, and latency (image capture ~1 s), with mitigations like clean surfaces, liveness checks, and proper thresholding [6] [7]. We provide a step-by-step Arduino integration guide: required parts, wiring, libraries (e.g. Adafruit_Fingerprint), and full example code for enrollment, verification, deletion and debugging. A sample fingerprint-controlled lock project is designed: parts list (with costs), wiring schematic, and code integrating the sensor with a servo or electric strike. A state-diagram charts the lock's logic (finger scan → match → unlock; fail→ retry or lockout). Finally, we give a troubleshooting checklist, testing tips and further references (books and tutorials). All claims and steps are documented with datasheets, vendor guides, and academic sources.

1. Common 4-Wire Optical Fingerprint Modules

Many Arduino-friendly optical modules use a 4-conductor cable (red, green, white, black) for power and UART. Notable examples include:

- **Zhian ZFM-20 series** (Hangzhou Zhian Tech, China): optical modules (AS601/AS608-based) storing 120–375 fingerprints. Official manual shows a 4-pin connector: Pin1=Vin (red), Pin2=TD out (green), Pin3=RD in (white), Pin4=GND (black) [1]. (Datasheet: Zhiantech *ZFM-20 Series Fingerprint Identification Module User Manual* v1.5).
- **AS608 / Adafruit Optical Module** (Adafruit 751, originally built by Zhian/SynoChip): captures 160×160 images with on-module DSP. Storage 162 templates. Supply 3.6–6 V [2]. Datasheet (HandsonTec/Robu) lists 120 mA, FRR<1% etc [2]. Adafruit's product page links to their datasheet and tutorial [8] [9].
- **R305/FPM10A** (ASR/GRDET?): an older optical module (product of Crossmatch/ASR) often sold by generic vendors. Typical specification: 4-pin (VCC, GND, TX, RX), storage ~100 templates, UART. Detailed manuals are scarce, but R305 clones follow the same wiring (see section 2).
- **GT-511/GT-521 series** (Goodix/ADH-Tech): optical sensors (258×202 px) with ARM MCU. Some variants (GT-521F32) use a 4-pin UART connector (VCC, GND, TX, RX) [10]. Goodix' GT-521 datasheet shows Pin4=VCC, Pin3=GND, Pin1=TX, Pin2=RX [10].
- **SparkFun/FPC2534 (R307S)**: SparkFun sells a Qwiic fingerprint (Optical) module (FPC2534) with I^2C/UART. It's based on the R307 sensor (Zhian ZFM-30). SparkFun's hookup guide and library cover it.

For each model, see the official manuals/datasheets for full specs: e.g. Zhian ZFM-20 manual [1], HandsonTec AS608 user guide [2], SparkFun Qwiic Fingerprint guide, etc. These show pinouts and capacities (table below).

Module (Manufacturer)	Wire Colors	Function (wire)	Notes / Specs	Source
Zhian **ZFM-20** (Hangzhou Zhian)	Red, Green, White, Black	Red=Vin, Green=TX (data out), White=RX (data in), Black=GND [1]	3.6–6V supply, 120–375 fingerprints (varies by model).	User manual [1]
AS608/Adafruit 751 (SynoChip/Adafruit)	Red, White, Green, Black	Red=VCC, Black=GND, White=RX, Green=TX [3]	3.6–6V, stores 162 templates [2]. Adafruit and clones.	Adafruit tutorial [3]; AS608 datasheet [2]
Generic **R305 (FPM10A)**	Red, Black, White, Green	Red=5V, Black=GND, White=RX, Green=TX (typical)**	~100 templates, 9600–57600 baud. (Wire colors as per many tutorials.)	Forum/ hardware guides (see text).
GT-521F32 (Goodix/ ADH)	Red, Green, White, Black	Red=VCC, Black=GND, White=RX, Green=TX (inferred)*	3.3–6V, up to 200 fingerprints [10].	Goodix datasheet [10]
SparkFun **Qwiic Fingerprint (FPC2534)**	Red, White, Green, Black?	Red=VCC, Black=GND, White=RX, Green=TX (UART mode).	Uses I^2C or UART (requires library).	SparkFun hook-up guide.

Note: Wire colors can vary by cable manufacturer. The convention above is most common [3] [1]. *Examples of R305 wiring (Uno firmware example): Red=VCC, Black=GND, Green=TXD, White=RXD. Actual assignment should be confirmed by the module's documentation or LED behavior.

2. Wiring Conventions (Red/Green/White/Black)

Across these models, the four wires almost always map as: **Red = VCC**, **Black = GND**, **White = RX (module's data in)**, **Green = TX (module's data out)** [3] [1]. The table below compares a few examples:

Module	Red (wire)	Green	White	Black	Notes
Zhian ZFM-20	Vin (Power) [1]	TD out (UART TX) [1]	RD in (UART RX) [1]	GND [1]	Cable color from official Zhian manual.

Module	Red (wire)	Green	White	Black	Notes
Adafruit AS608	+V (3.6–6V) [3]	UART TX (to MCU) [3]	UART RX (from MCU) [3]	GND [3]	Adafruit wiring guide for “slim connector”.
SparkFun (UART)	+V (5V)	TXD (module out)	RXD (module in)	GND	SparkFun FPC2534 uses same convention (via library).
Generic R30x	+5V	TX	RX	GND	Many R305/R307 modules follow Red=Vcc, Black=GND, White=RX, Green=TX (as community reports).
GT-521F32 (Goodix)	VCC [10]	TX (to MCU) [10]	RX (from MCU) [10]	GND [10]	Goodix datasheet: Pin4=VCC, Pin1=TX, Pin2=RX, Pin3=GND [10].

This shows a consistent pattern: **Red=power, Black=ground**, with **White/Green swapped between RX/TX** depending on model and orientation. (Always double-check the module’s documentation!) For example, Zhian’s pinouts explicitly label “TD out (green)” as module→Arduino and “RD in (white)” as Arduino→module [1]. Adafruit’s guide confirms white=RX, green=TX [3].

3. Optical Fingerprint Sensor Principle

Optical sensors image the fingerprint by shining light onto the fingertip and capturing the reflected pattern. A finger is placed on the sensor’s transparent window; LEDs illuminate the ridges and valleys differently (ridges absorb more light) [4]. An internal CMOS/CCD camera produces a grayscale image of the print. Onboard DSP logic then **processes** the image: it may apply contrast enhancement and extract *minutiae* (ridge endings, bifurcations), encoding them as a template [5]. For enrollment, the template is saved in flash memory. For matching, a new scan’s template is compared against stored templates using a matching algorithm (often 1:1 or 1:N matching). If similarity exceeds a threshold, a fingerprint is recognized [5]. These steps (capture, processing, matching) occur entirely in the module; the Arduino communicates via serial commands to enroll or search.

Figure: A smartphone with integrated fingerprint reader (symbolic of optical scanning). In modules, LEDs illuminate the finger and a small camera/sensor captures the ridge patterns.

```
flowchart LR
    Finger[Fingertip on Window] --> Illumination[LED Illumination]
    Illumination --> ImageCapture[Sensor (camera) captures image]
    ImageCapture --> ImageProc[Image Processing\n(+ enhancement)]
    ImageProc --> FeatureExtract[Feature Extraction (minutiae)]
    FeatureExtract --> TemplateMatch[Template Comparison]
    TemplateMatch -->|Match?| Match{Similarity above\nthreshold?}
    Match -->|Yes| Success[Fingerprint Identified]
    Match -->|No| Reject[No Match (reject)]
    TemplateMatch -->|Create Template| Store[Store in Memory]
```

- **Fingerprint imaging:** Light (LED) shines through/around the finger; valleys reflect more light into the sensor, ridges reflect less, producing a high-contrast grayscale image [4].
- **Image processing:** The raw image is digitized. Preprocessing (noise reduction, normalization) readies it for feature detection.
- **Feature extraction:** Key points (minutiae) are extracted. These form a compact "signature" (each ~512 bytes for a template).
- **Storage/Matching:** Templates are stored in onboard flash (e.g. 162 slots for AS608 [2]). A matching algorithm (based on distance of minutiae or pattern matching) compares the scanned template to saved ones [5]. A result (ID or failure) is returned.

4. Limitations & Failure Modes

Optical fingerprint modules work well but have known limitations:

- **Environmental / Finger Condition:** Dirt, grease, or moisture on the finger or sensor can degrade image quality. Damaged or extremely dry skin also yields poor scans [6]. Bright ambient light or very cold fingers may cause misses. *Mitigation:* Prompt users to clean the glass and finger; use multiple scans or higher exposure; keep sensors in controlled enclosure.
- **Spoofing (Presentation Attacks):** Glassy sensors can be fooled by fake fingerprints (e.g. gummy replicas). Traditional models lack liveness detection [7]. *Mitigation:* Use active anti-spoof methods (multi-spectral imaging, pulse oximetry) if available; require user interaction (press/lick finger); combine with PIN.
- **Wear and Tear:** The protective window can scratch or the LEDs/camera can degrade over time. This reduces image contrast. *Mitigation:* Use tempered glass or clear plastic covers; design module so cover can be replaced; periodically recalibrate or adjust thresholds.
- **False Accept/Reject Rates:** Even at security level 3, AS608's FAR is <0.001% and FRR <1% [2]. A nonzero FAR means rare false unlocks; FRR means occasionally a genuine finger is rejected. *Mitigation:* Tune security level/thresholds; allow multiple swipes; use multi-factor verification for high-security.
- **Throughput (Speed):** Image capture and matching typically take ~0.5–1 s per fingerprint. Bulk enrollment or search over many templates can be slow. *Mitigation:* Use lower baud rates only if signal issues; ensure no blocking code; use the fastest allowed baud (57600 default) and filter interrupts properly.
- **Memory Limits:** Modules have fixed template capacity (e.g. 120/162 slots [2]). Once full, no new enrollments are possible. *Mitigation:* Provide admin tools to delete old templates; periodically back up templates; consider multiple modules or external flash if a large database is needed.

Each of these issues can be managed by system design: keeping the sensor clean, using anti-spoofing strategies, validating performance (false rate) in your context, and having an administrative override (e.g. PIN code) in case of lockouts.

5. Arduino Integration Guide

Parts Needed:

- **Fingerprint sensor module** (e.g. AS608/ZFM-20). Price: \$15–50.
- **Arduino board** (Uno/Nano/Mega, etc). Price: \$10–25. Note: Use hardware Serial1 on Mega/Nano 33, or SoftwareSerial on Uno/Nano for the sensor's UART.
- **Connecting wires**: 4 jumper wires (female/female or Dupont cable).
- (Optional) **Level shifter** if module requires 3.3 V logic (many work at 5 V).
- **Breadboard / prototyping board** for easy wiring.
- **12 V power supply / battery** (if using servo or solenoid lock). Arduino itself can be USB-powered.
- **Output device**: e.g. small servo (~\$5) or 12 V electric strike/solenoid (~\$10–15) for a lock.
- **Misc:** Pull-up/down resistors if needed for servo control, connectors, etc.

Wiring Steps:
1. **Power:** Connect the module's Red wire to Arduino +5 V (or 3.3 V if specified), and Black to GND. Some

modules accept 5 V but use on-board regulator. Verify module voltage (Adafruit AS608 wants 5 V in, outputs 3.3 V internally [9]).
2. **UART:** Connect module TX (Green) to Arduino RX pin, and module RX (White) to Arduino TX pin. *On an Uno/Nano*: if using SoftwareSerial, pick two digital pins and swap: e.g. connect Green→pin2 (SW Rx) and White→pin3 (SW Tx). If using hardware Serial (Mega), connect directly to RX1/TX1. Ensure a common ground.
3. **Indicator LED (optional):** Modules often have an onboard red LED that flashes on scan. Use it as "finger detected" indicator (it lights when finger is on). No extra wiring needed unless you add an external LED for status.
4. **Output Actuator:** Wire servo or lock coil to Arduino digital pin (e.g. pin9) and to external power. **Do not** power a servo or strike from Arduino 5 V pin if it draws >200 mA; use a separate supply and transistor/MOSFET driver. Return the grounds all together.

Libraries:

- **Adafruit Fingerprint library** ("Adafruit_Fingerprint"). Supports enrollment, search, delete, read image, etc [8]. Install via Library Manager.
- **SparkFun FPC2534 library**: If using SparkFun's I^2C/UART modules, their library (FPC2534) is required.
- **Other forks:** The Adafruit library itself is on GitHub [11]. Some use "Fingerprint2" library from GitHub (public domain).
- **Arduino code examples:** Both Adafruit and SparkFun libraries include example sketches: `Enroll.ino`, `Fingerprint.ino`, `Search.ino`, `Delete.ino`, etc. These demonstrate how to capture, store, and match prints. Adafruit's tutorial instructs running *File→Examples→Adafruit_Fingerprint→enroll* [12].

Example Functions: (pseudocode outline)

- **EnrollFinger(id):** Prompt user, capture image twice, call `GenChar`, `RegModel`, `StoreChar` (Adafruit library) to save under an ID.
- **VerifyFinger():** Call `getImage()` and `image2Tz()`, then `search()` to check against stored IDs. Returns matched ID or -1.
- **DeleteUser(id):** Use `deleteModel(id)` command to remove a template.

The sketch below is a self-contained example (simplified and annotated) for Arduino, showing setup and loop for enrollment, search, and debug output:

```
#include <Adafruit_Fingerprint.h>
#include <SoftwareSerial.h>

// Connect sensor TX (Green) to Arduino pin 2, sensor RX (White) to pin 3
SoftwareSerial fingSerial(2, 3);
Adafruit_Fingerprint finger = Adafruit_Fingerprint(&fingSerial);

void setup() {
  Serial.begin(115200);
  while (!Serial);  // wait for serial monitor
  Serial.println("Fingerprint Sensor Test");
```

```
  finger.begin(57600);  // Sensor baud rate
  if (finger.verifyPassword()) {
    Serial.println("Sensor found!");
  } else {
    Serial.println("Sensor not found :(");
    while (1);
  }
}

void loop() {
  // Check if a finger is available
  if (finger.getImage() != FINGERPRINT_OK) {
    return; // no finger found
  }
  // Convert to characteristics
  finger.image2Tz(1);

  // Search against stored templates
  uint8_t p = finger.fingerSearch();
  if (p == FINGERPRINT_OK) {
    Serial.print("Found ID #"); Serial.print(finger.fingerID);
    Serial.print(" with confidence "); Serial.println(finger.confidence);
    // e.g., trigger door unlock here (not shown)
  } else {
    Serial.println("No match");
  }

  delay(1000); // Debounce time
}

// Helper: enroll a new fingerprint into slot 'id'
void enrollFingerprint(uint8_t id) {
  Serial.print("Enrolling ID #"); Serial.println(id);
  while (finger.getImage() != FINGERPRINT_OK);
  finger.image2Tz(1);
  Serial.println("Remove finger");
  delay(2000);
  while (finger.getImage() != FINGERPRINT_OK);
  finger.image2Tz(2);
  int ok = finger.createModel();
  if (ok == FINGERPRINT_OK) {
    finger.storeModel(id);
    Serial.println("Stored!");
  } else {
    Serial.println("Failed to create model");
  }
}
```

Annotations: This sketch initializes the sensor, captures a finger image, and performs a 1:N search. The `enrollFingerprint(id)` routine (commented) shows how to enroll. All debug output goes to the Serial Monitor. Adjust pins and power lines to your hardware (e.g. use `Serial1` on Mega instead of SoftwareSerial).

6. Sample Fingerprint-Access Lock Project

Parts List (approx. costs):

- Fingerprint sensor module (AS608/ZFM-20): \$20–\$40.
- Arduino Uno or Nano: \$10–\$20. (Mega \$15 if needed.)
- Breadboard & jumpers: \$5.
- Servo motor (~1–2 kg·cm) or **12 V solenoid strike**: \$5–\$15.
- 12 V DC power supply (for solenoid) or 5–6 V regulator for servo: \$10.
- (Optional) Passive buzzer or LED for status: \$1.
- Enclosure materials, mounting hardware: \$10.

Wiring Schematic:
1. Connect sensor to Arduino as above (Vcc, GND, RX/TX).
2. Connect servo: Arduino digital pin → signal; servo Vcc to 5V (or external regulated 5V), GND common with Arduino. Use a transistor/MOSFET if driving a larger load.
3. If using a 12 V lock: connect lock coil to +12 V supply and to Arduino output through a MOSFET switch (Arduino pin drives gate). Include flyback diode. Grounds all tied.
4. (Optional) Connect a push-button for admin reset to another input pin with pull-up.

Example Lock Code Integration:
Extend the earlier sketch by adding lock control and attempt tracking. Pseudocode:

```
const int lockPin = 9;      // Servo or MOSFET gate pin
int attemptCount = 0;
const int MAX_ATTEMPTS = 3;
unsigned long unlockTimer = 0;

void setup() {
  // (Initialize sensor and serial as before)
  pinMode(lockPin, OUTPUT);
  digitalWrite(lockPin, LOW); // Locked by default
}

void loop() {
  if (finger.getImage() == FINGERPRINT_OK) {
    finger.image2Tz(1);
    if (finger.fingerSearch() == FINGERPRINT_OK) {
      Serial.println("Access GRANTED");
      digitalWrite(lockPin, HIGH); // Unlock (activate servo/strike)
      unlockTimer = millis();
```

```
      attemptCount = 0;
    } else {
      Serial.println("Access DENIED");
      attemptCount++;
      if (attemptCount >= MAX_ATTEMPTS) {
        Serial.println("LOCKOUT - too many attempts");
        // Optionally implement lockout (ignore fingers for a timeout)
      }
    }
    delay(1000);
  }
  // Relock after 5 seconds
  if (unlockTimer > 0 && millis() - unlockTimer > 5000) {
    digitalWrite(lockPin, LOW);
    unlockTimer = 0;
    Serial.println("Relocked");
  }
}
```

Lockout & Safety: After *N* failed scans (e.g. 3), the system can enter a *lockout* state requiring a master reset. In our design, an optional keypad button or magnetic key can reset `attemptCount`. Always ensure the lock mechanism fails safe (e.g. remains locked on power loss if needed). Never draw heavy lock current through Arduino pins directly – use appropriate drivers.

```
stateDiagram-v2
    [*] --> Idle
    Idle --> Scanning : Finger placed
    Scanning --> Verify
    Verify --> Unlocked : Match (finger OK)
    Verify --> Error : No match
    Unlocked --> Idle : Timeout/Key removed
    Error --> Idle : (if <3 tries) Reset scan
    Error --> Lockout : (>=3 tries) Lockout state
    Lockout --> Idle : Master reset (button)
```

7. Troubleshooting & Testing

- **Power Check:** Ensure correct voltage. Many modules accept 5 V in (they have on-board regulators), but verify if your sensor is 3.3 V-only. Check that Arduino ground and sensor ground are common.
- **Wiring / Serial:** Double-check RX/TX. Remember RX ↔ TX swap. If using SoftwareSerial on Uno, use pins that support change interrupts (2,3 usually). Ensure baud matches (default 57600 for many modules [9]).
- **LED Indicator:** On power-up, the sensor's LED often blinks. When placing a finger, it should illuminate steadily. No LED means no power or sensor failure.

- **Library Version:** Use the latest Adafruit_Fingerprint or SparkFun library. In code, call `finger.begin(baud)` and `finger.verifyPassword()` to test connectivity (it returns FINGERPRINT_OK if module responds) [12].
- **Serial Debug:** Enable Serial.prints (as above) to see feedback. If you get "Sensor not found," check wiring or try another serial port.
- **Enroll Test:** Use the library's example *enroll* sketch on one finger. Then use *fingerprint* or custom code to verify you can match it. If enrollment fails, check if the finger was kept steady and clean during capture (module needs two consistent scans).
- **Ghosting / False Prints:** Wipe the sensor glass between tests. Slight ghost images can stay momentarily.
- **SerialMonitor:** Avoid very high baud at first. Start with 9600 and test, then increase.
- **Cross-talk:** If using SoftwareSerial, ensure no other code blocks during `finger.getImage()`. If problems persist, try Hardware Serial on Mega/Nano 33 as they have multiple UARTs.

8. Further Resources

- **Tutorials/GitHub:** Adafruit's learning guide has detailed instructions and example code [3] [12]. SparkFun's documentation covers their Qwiic fingerprint module (FPC2534) and library [13].
- **Books & Papers:** *Handbook of Fingerprint Recognition* (Maltoni et al.) is an authoritative text on fingerprint imaging and matching [14]. Academic surveys on biometrics (e.g. Jain's publications) discuss sensor technologies and security.
- **Reference APIs:** The Adafruit_Fingerprint library's GitHub (Adafruit/Adafruit-Fingerprint-Sensor-Library) and example sketches show all sensor commands. Also see SparkFun FPC2534 Arduino library.
- **Community:** Arduino forums and StackExchange often have solved Q&As for common issues (e.g. sensor wiring, model differences). Technical blogs (e.g. RandomNerdTutorials) offer step-by-step guides.
- **Related Technologies:** For broader context, compare with capacitive fingerprint sensors (e.g. GT-511C3 series) via Seeed Wiki or vendor docs to understand pros/cons. Standards on biometric performance (NIST FRR/FAR guidelines) can inform reliability expectations.

Sources: Manufacturer manuals and credible tutorials have been cited throughout. For example, the Zhian ZFM-20 datasheet [1] gives pinouts; Adafruit's product guide [8] provides specifications; a fingerprint security article [4] [5] explains optics and processing; and Adafruit's tutorial [12] shows Arduino integration. All code and claims here are grounded in those sources.

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